



Nondegenerate n-type doping phenomenon on molybdenum disulfide (MoS₂) by zinc oxide (ZnO)



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ABSTRACT

In this paper, we have demonstrated nondegenerate n-type doping phenomenon of MoS₂ by ZnO. The ZnO doping effects were systematically investigated by Raman spectroscopy and electrical/optical measurements (I_D - V_G with/without exposure to 520, 655, 785, and 850 nm laser sources). The ZnO doping improved the performance parameters of MoS₂-based electronics (I_{on} , μ_{FE} , n) owing to reduction of the effective barrier height between the source and the MoS₂ channel. We also monitored the effects of ZnO doping during exposure to air; reduction in ΔV_{TH} of about 75% was observed after 156 h. In addition, the optoelectronic performance of the MoS₂ photodetector was enhanced due to the reduction of the recombination rate of photogenerated carriers caused by ZnO doping. In our results, the highest photoresponsivity (about 3.18×10^3 A/W) and detectivity (5.94×10^{12} Jones) of the ZnO-doped photodetector were observed for 520 nm laser exposure.

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1. Introduction

Transition metal dichalcogenides (TMDs) with two-dimensional (2D) semiconducting layered structure, such as molybdenum disulfide (MoS₂) and tungsten diselenide (WSe₂), have been considered to be extremely promising materials for future transparent, flexible and wearable electronic/optoelectronic devices because of their exceptional electrical [1–3], optoelectronic [4] and mechanical properties [5]. Because of the excellent thickness scalability of TMDs as an atomic monolayer and van der Waals epitaxial structure without dangling bonds, TMD-based electronic devices have immunity to the short-channel effects; thus, they have been contemplated for future high-performance nanoelectronic applications [2,6]. In addition, TMD materials are expected to be suitable for high-performance optoelectronic devices (i.e., photodetectors and solar cells) because of their tunable band gap (between 1.2 and 1.8 eV, depending on layer thickness) and extremely high quantum efficiency [7,8].

To further improve the performance of TMD-based electronic/optoelectronic devices, it is necessary to reduce the contact resistance of metal–semiconductor junctions and to lower the recombination rate of carriers. However, such improvements cannot be carried out by using the ion implantation technique on TMD material because of the crystal damage of the 2D structure. For these reasons, recent work has focused on the developing of less disruptive doping techniques for application to TMD devices. Recently, Fang et al. reported p- and n-type doping technique on TMD devices, which involved the use of NO₂ molecules and potassium, respectively [9,10]. However, both p- and n-doping techniques produced very high levels of doping, making the TMD channel layers near metallic. Thus, development of lighter doping techniques for use with TMD devices is crucial for the optimization of TMD-based electronics/optoelectronics. Lin et al. and Park et al. reported low-level doping techniques using Cs₂CO₃ ($\sim 1 \times 10^{11}$ cm⁻² on MoS₂) and DNA/M-DNA ($\sim 6.4 \times 10^{10}$ cm⁻² on MoS₂), respectively [11,12].

Here, we also reported the nondegenerate n-type doping technique on MoS₂ by zinc oxide (ZnO) film coating. The negative polar surface of the ZnO film was expected to reduce the hole carrier density in the MoS₂ channel layer, thereby causing an n-doping effect. We investigated the ZnO doping effects on MoS₂ by Raman spectroscopy and electrical measurements (I_D - V_G and I_D - V_D). In addition, we also monitored the V_{TH} variations of ZnO-doped MoS₂ devices during 156 h of exposure to air. Finally, the

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effects of ZnO doping upon the photoresponsivity and detectivity performance of MoS₂-based optoelectronic devices were observed by means of optical measurements performed under exposure to laser sources of wavelengths 520, 655, 785, and 850 nm.

2. Experiments

For the fabrication of back-gated MoS₂ devices (transistors and photodetectors), source/drain electrode regions were patterned (channel length and width are 5 μm) on MoS₂/SiO₂/Si samples by optical lithography, followed by Ti (10 nm) and Au (50 nm) deposition via e-beam evaporation. ZnO solutions were prepared by dissolving 1.1 g of zinc acetate dihydrate [Zn(CH₃COO)₂·2H₂O] and 0.3 mL of monoethanolamine (MEA) in 8.96 mL of 2-methoxyethanol as the solute, stabilizer and solvent, respectively. The molar ratio of MEA to zinc acetate dihydrate was maintained at 1.0 and the concentration of zinc acetate dihydrate was 0.5 M. The solution was stirred at 60 °C for 2 h in air to obtain a transparent and homogeneous solution, followed by aging at room temperature for 5 days. For doping of MoS₂ devices, the ZnO solutions were dropped onto MoS₂ devices and spin coated at 3000 rpm for 15 s. Subsequently, the ZnO-doped MoS₂ devices were annealed by using a thermal furnace operated at 250 °C for 15 min in air ambient. The devices were electrically analyzed by using an HP4155A semiconductor parameter analyzer, and the carrier concentration (n), field-effect mobility (μ_{FE}) and threshold voltage (V_{TH}) were calculated by using the current–voltage ($I_{\text{D}}-V_{\text{G}}$) data. To investigate the optoelectronic properties of the fabricated ZnO-doped MoS₂ devices, the $I_{\text{D}}-V_{\text{G}}$ measurements were performed under both dark and illuminated conditions. Here, all drain currents (I_{D}) were normalized by channel width (W). The light sources were set up by using a dot laser of wavelength 520, 655, 785, and 850 nm and an optical power of 6 mW/cm². Photoresponsivity (R) and detectivity (D^*) were calculated by using the acquired $I_{\text{D}}-V_{\text{G}}$ curves.

3. Results and discussion

To verify the doping effects of ZnO film on MoS₂ film, we performed Raman spectroscopy on ZnO-doped MoS₂ samples exfoliated on a SiO₂/Si substrate. Fig. 1a shows Raman spectra of undoped/ZnO-doped MoS₂ samples. In MoS₂ samples, the two conventional peaks (E_{12g} and A_{1g}) were observed at about 380 and 406 cm⁻¹, attributed to the in-plane (E_{12g} peak) and out-of-plane

(A_{1g} peak) vibrations of MoS₂ [13]. We then extracted E_{12g} and A_{1g} peak positions for each undoped/doped MoS₂ sample (Fig. 1b). In the case of control samples of undoped MoS₂, the E_{12g} and A_{1g} peak positions were located at 380–380.5 and 405.2–405.8 cm⁻¹, respectively. After ZnO doping, those two peaks were red-shifted (379.4–380.2 cm⁻¹ for E_{12g} and 404.6–405.3 cm⁻¹ for A_{1g}). These phenomena indicated that ZnO causes the softening of MoS₂ vibrational modes and consequently causes n-type doping effects in MoS₂ films [14]. The Raman shifts of the two peaks were 0.3–0.6 cm⁻¹, indicating that the ZnO treatment produced low levels of doping, as desired.

To confirm the doping effects of ZnO on MoS₂ once again, we fabricated MoS₂-based transistors and performed the electrical measurement ($I_{\text{D}}-V_{\text{G}}$). Fig. 2a shows the three-dimensional schematic illustrations of ZnO-doped MoS₂ back-gated transistors and an energy band diagram of the metal (Ti)-undoped/doped MoS₂ junctions. Because the ZnO crystal structure is asymmetric wurtzite, it has ionic and polar structures consisting of oxygen- and zinc-terminated surfaces [15]. From these effects, the hole carriers further accumulated at the interface between ZnO and MoS₂, thereby causing the n-doping phenomenon on the MoS₂ channel layer. As shown in Fig. 2a, these negative polar surface of ZnO was expected to be reduced the effective barrier height of the Ti-MoS₂ junction by shifting down of the MoS₂ energy band at the source–MoS₂ junction (Schottky barrier lowering effect: $\Phi_{\text{Control,eff}} > \Phi_{\text{ZnO,eff}}$), eventually influencing the tunneling of electron carriers from the source metal to MoS₂. This ZnO doping mechanism is similar to that of self-assembled monolayer (SAM) [16,17] and DNA/M-DNA doping [12]. As a result, a negative shift in threshold voltage (V_{TH}) from −3.8 to −11.6 V and an on-current enhancement from 1.6×10^{-6} to 8.27×10^{-6} A/ μm were observed in the ZnO-doped MoS₂ transistors (Fig. 2b). We then calculated field-effect mobility (μ_{FE}) and 2D sheet concentrations (n) of undoped/ZnO-doped MoS₂ devices for investigating the changes in electronic performance caused by ZnO doping. Here, the field-effect mobility and concentration were extracted by using the following equations: $\mu_{\text{FE}} = L/(WV_{\text{D}}C_{\text{OX}}) \times (\partial I_{\text{D}}/\partial V_{\text{G}})$ and $n = I_{\text{D}}L/qW\mu_{\text{FE}}V_{\text{D}}$, where q is the electron charge, L and W are the length and width of the channel, respectively, and C_{OX} is the gate oxide capacitance per unit area, $\epsilon_{\text{OX}} \times \epsilon_0/t_{\text{OX}}$. As shown in Fig. 2(c), the field-effect mobility value was increased from 12.8 to 29.3 cm²/V-sec after ZnO doping, indicating that the ZnO doping improved the field-effect mobility by increasing the carrier injection probability. This mobility improvement phenomenon by doping has also been reported

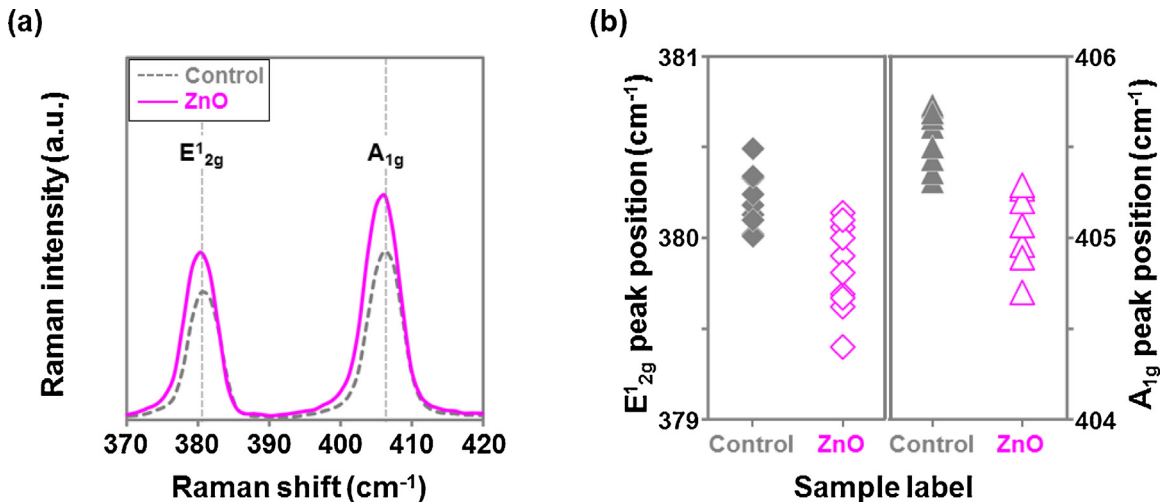


Fig. 1. (a) Raman spectra of undoped/ZnO-doped MoS₂ films. (b) Extracted E_{12g} and A_{1g} peak positions of undoped/ZnO-doped MoS₂ films.

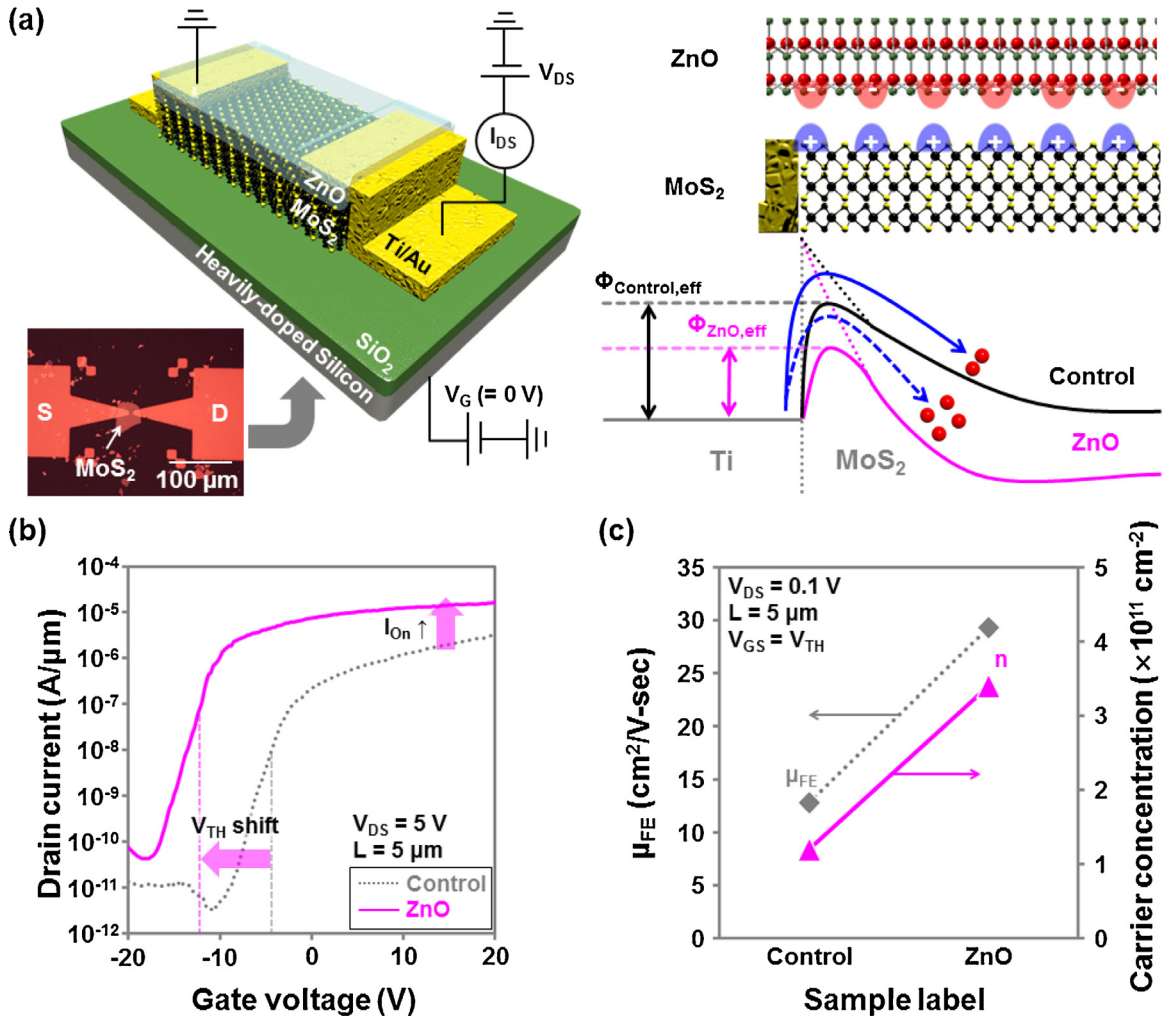


Fig. 2. (a) Three-dimensional schematic illustration and OM image of back-gated ZnO-doped MoS₂ transistor and energy band diagram of source (Ti)–MoS₂ junction. (b) Electrical characterization of MoS₂ transistor doped by ZnO film. (c) Carrier concentration (n) and field-effect mobility (μ_{FE}) of undoped/ZnO-doped MoS₂ transistor. Here, the I_D was normalized by channel width (W).

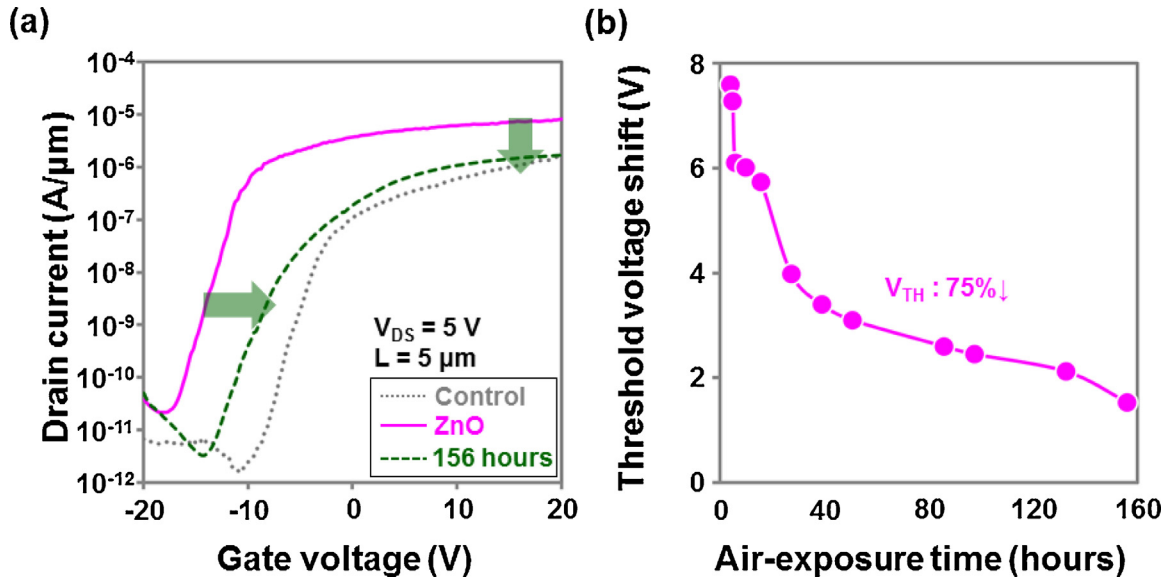


Fig. 3. (a) I_D – V_G characteristics of a ZnO-doped MoS₂ device before and after about 156 h of exposure to air. (b) Threshold voltage shift ($\Delta V_{TH} = V_{TH,ZnO} - V_{TH,Control}$) of ZnO-doped MoS₂ transistor as a function of air exposure time.

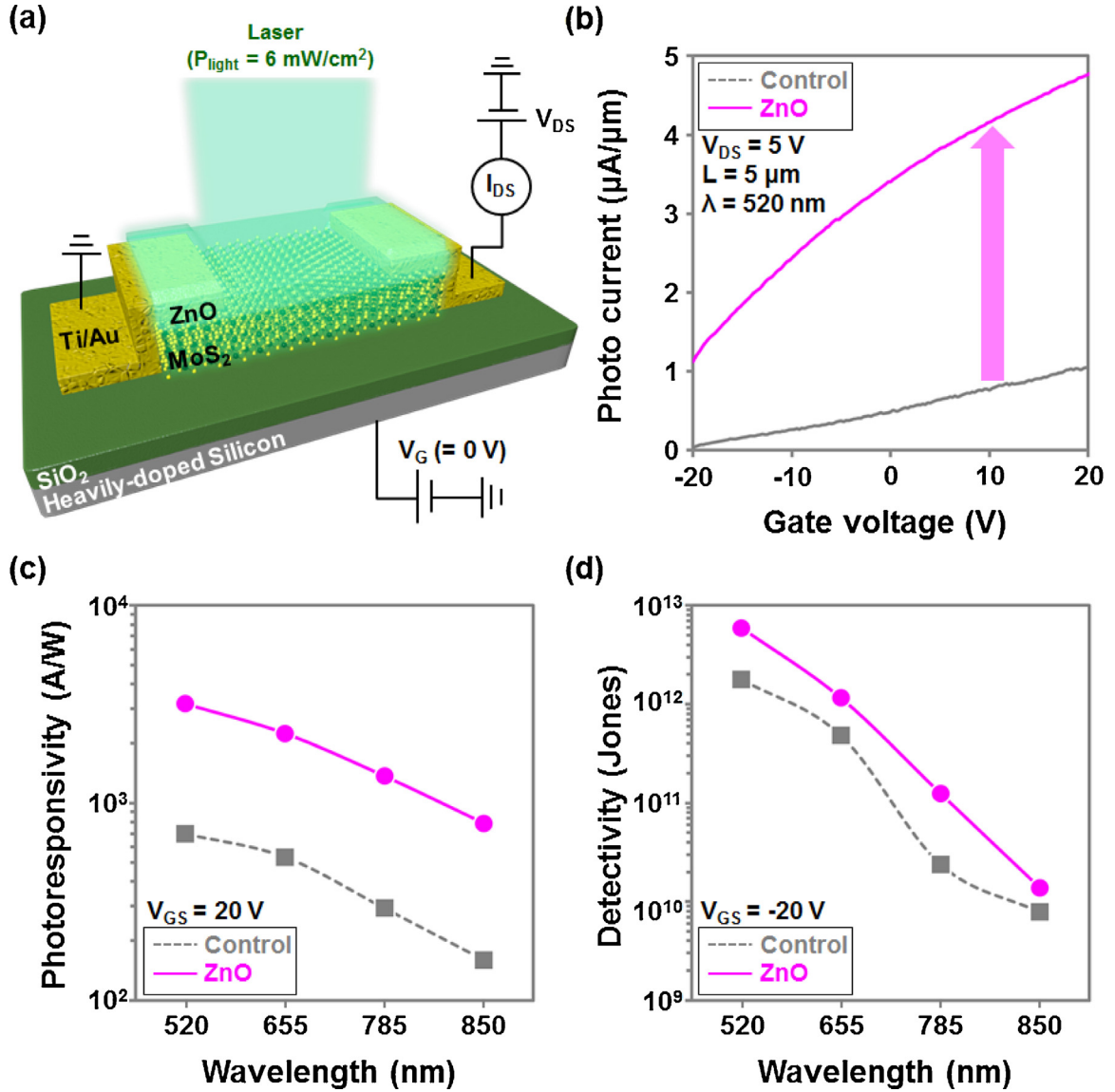


Fig. 4. (a) Schematic diagrams of the ZnO-doped MoS₂ photodetector under exposure to a light source ($P_{\text{light}} = 6 \text{ mW/cm}^2$). (b) Photocurrent, (c) extracted photoresponsivity, and (d) extracted detectivity of the ZnO-doped MoS₂ photodetector under exposure to a 520 nm laser source.

recently by other researchers [12,16]. In addition, the 2D sheet doping concentration of MoS₂ transistors was also increased by ZnO doping from 1.22×10^{11} to $3.41 \times 10^{11} \text{ cm}^{-2}$. Based on this result, the ZnO doping was expected to be of the non-degenerate doping regime, similar to DNA n-doping ($\sim 6.4 \times 10^{10} \text{ cm}^{-2}$) and OTS p-doping ($\sim 1.4 \times 10^{11} \text{ cm}^{-2}$) [12,17].

In order to investigate the stability of the ZnO doping effects, we measured I_D - V_G characteristics and V_{TH} values of the ZnO-doped MoS₂ transistors during 156 h exposure to air. In the ZnO-doped MoS₂ device, we observed an on-current reduction (from 8.27×10^{-6} to $1.71 \times 10^{-6} \text{ A/}\mu\text{m}$) and a threshold voltage shift (-11.6 to -4.7 V) after 156 h of air exposure (Fig. 3a), which indicated that the ZnO n-doping effect was weakened by exposure to moisture. When H₂O molecules were adsorbed onto the ZnO film surface, they partially dissociated into H⁺ and OH⁻ species and H⁺ ions were diffused into the ZnO film, subsequently making strong bonds with the anion (O²⁻) of ZnO [18]. This bonding seemed to weaken the dipole moments of the negative polar surfaces of ZnO, thereby weakening the n-doping phenomenon. We extracted V_{TH} values for ZnO-doped MoS₂ devices during the exposure to air; ΔV_{TH} steadily decreased with increasing air

exposure time. After 156 h, the ΔV_{TH} value was decreased by about 75.6% (from 7.8 to 1.8 V).

Finally, we investigated the effects of ZnO doping on optoelectronic device performance by performing an electrical measurement (I_D - V_G) of a ZnO-doped MoS₂ photodetector device using a laser source of various wavelengths (520, 655, 785 and 850 nm). Fig. 4a shows a schematic diagram of a ZnO-doped MoS₂ photodetector. Here, the incident power of the laser source was 6 mW/cm^2 . When the ZnO-doped MoS₂ photodetector was exposed to the laser source, the excited electron-hole pairs were expected to form more tightly bound trions due to the increase of electron density in MoS₂ [19], consequently reducing the recombination rate of photoexcited carriers in the ZnO-doped MoS₂ photodetector. We clearly observed that the photocurrent through all gate regions was increased compared to the control sample during a voltage sweep from -20 to 20 V (from 4.01×10^{-8} to $1.12 \times 10^{-6} \text{ A/}\mu\text{m}$ at -20 V , and from 1.05×10^{-6} to $4.76 \times 10^{-6} \text{ A/}\mu\text{m}$ at 20 V , as shown in Fig. 4b). We then extracted the photoresponsivity ($R = I_{\text{photo}}/P_{\text{light}}$ at $V_{\text{GS}} = 20 \text{ V}$) and detectivity ($D^* = (RA^{1/2})/(2eI_{\text{dark}})^{1/2}$ at $V_{\text{GS}} = -20 \text{ V}$) by using various laser sources (520, 655, 785, and 850 nm, as shown in Fig. 4c-d). In the case

of the 520 nm wavelength, the photoresponsivity was increased by a factor of about 4.55 (from 6.98×10^2 to 3.18×10^3 A/W) and the detectivity was increased by a factor of 3.34 (from 1.78×10^{12} to 5.94×10^{12} Jones). The photoresponsivity of the ZnO-doped MoS₂ photodetector was higher than that recently reported for other TMD-based photodetectors [4,20]. In addition, photoresponsivity and detectivity were clearly observed at all wavelengths tested, not only in response to the 520 nm laser source, because the decreased electron–hole recombination helped to collect more photogenerated electrons. We also note that the photoresponsivity and detectivity were increased for the shorter wavelengths ($R_{520\text{nm}} > R_{850\text{nm}}$ and $D_{520\text{nm}}^+ > D_{850\text{nm}}^+$), due to the generation of more photocarriers by higher-energy photons [16] and the higher absorption of the MoS₂ film [21].

4. Conclusion

In conclusion, we reported nondegenerate n-doping technique for MoS₂ by ZnO film. An oxygen-terminated polar surface of ZnO caused hole carriers to accumulate at its interface with the MoS₂ channel, adjusting the performance parameters of MoS₂ electronic devices (with increases in I_{on} , μ_{FE} , and n , as well as a shift in V_{TH}) due to a reduction in the effective barrier height between the source and the MoS₂ channel. We monitored the effects of ZnO doping during exposure to air, observing a reduction in ΔV_{TH} values of about 75% after 156 h. In addition, we investigated the optoelectronic performance improvements of MoS₂ photodetectors employing ZnO doping; the highest photoresponsivity ($\sim 3.18 \times 10^3$ A/W) and detectivity (5.94×10^{12} Jones) were observed under exposure to a 520 nm laser source.

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